# RESUME

## PERSONAL PROFILE

Name: Muhammad Iqbal, Shamreed

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Date of Birth: 22/08/2004

Place of Birth: Surathkal, Karnataka

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GitHub: https://github.com/Shamreed

Portfolio: https://shamreed.github.io/Shamreed-portfolio/

## EDUCATION:

2022- 2026 Bachelor of Technology in Artificial Intelligence and Machine Learning
NMAM Institute of Technology, Nitte, Karkala, India
Affiliated to Nitte University.

CGPA: 8.66

## PROJECTS:

**Pothole Detection and Severity Analysis** | Python, YOLOv8, PyTorch, Django, React

- Developed a deep learning - based pothole detection and severity analysis system using YOLOv8 on dashcam video data.
- Built an end-to-end processing pipeline integrating OCR, GPS data extraction, and database storage for geospatial pothole mapping.
- Developed a Django-React web application to visualize detected potholes on an interactive map, enabling efficient monitoring and analysis.

**Gear-Driven Walking Robot** | Mechanical Design, Gears, Linkages, Motors, Robotics, Hardware Integration

- Designed and built a compact walking robot using interconnected gears, mechanical linkages, and a motor-driven locomotion mechanism.
- Developed and assembled a gear-based actuation system that converts motor rotation into coordinated leg movement for stable walking motion.
- Integrated a rechargeable battery and solar panel for dual-power operation, and tested the mechanical system for reliable locomotion and overall performance.

**Alzheimer's Disease Classification Using Deep Learning** | Python, TensorFlow, Keras, Deep Learning, Transfer Learning, CNNs

- Developed a deep learning-based image classification system for Alzheimer's disease classification.
- Trained and evaluated multiple CNN architectures, including DenseNet169, VGG16, ResNet50, and EfficientNetB0, using transfer learning.
- Conducted model comparison and performance evaluation using accuracy, precision, recall, F1-score, and ROC-AUC to identify the most effective architecture.

## TECHNICAL SKILLS:

- **Programming Languages:** Python, C++, Java, JavaScript, SQL
- **Machine Learning & Deep Learning:** Machine Learning, Deep Learning, Neural Networks, Transfer Learning, Model Training & Fine-Tuning, Model Evaluation, Reinforcement Learning, PyTorch, TensorFlow, Keras, scikit-learn
- **Multimodal AI & Robotics:** Multimodal AI, ROS 2 (Basic), Arduino, Hardware Integration
- **Software Engineering:** Git, GitHub, Docker, REST APIs, Spring Boot
- **Databases & Web:** MySQL, React.js, HTML, CSS
- **Hardware & Manufacturing:** Soldering, Basic Electrical Wiring, Quality Inspection

## INTERNSHIPS:

**Software Development Intern - Jun 2025 - Aug 2025**

**IHUB IT Solutions Mangalore, Karnataka**

- Developed backend services for a full-stack Blog platform using Java Spring Boot, implementing secure REST APIs and role-based authentication.
- Integrated MySQL databases and implemented role-based authentication to support secure and scalable application workflows.
- Followed Git-based software engineering practices, contributing to debugging, testing, code maintainability, and reliable system development.

## LANGUAGE PROFICIENCY:

- Japanese Language (N5): (Completed NPTEL/SWAYAM course)
- English: Full professional proficiency
- Hindi: Full professional proficiency
- Kannada: Full professional proficiency
- Tulu: Full professional proficiency
- Byari: Full professional proficiency

## ACHIEVEMENTS:

- **2nd Place – Final Year Major Project Expo:** Recognized for the development of the *Pothole Detection and Severity Analysis* system, a multimodal AI solution integrating computer vision, depth estimation, and sensor data.
- **Certificate of Presentation – 7th International Conference on Computational Vision and Bio-Inspired Computing (ICCVBIC 2026):** Presented the *Pothole Detection and Severity Analysis* project at an international technical conference.

## CERTIFICATIONS:

- Introduction to Japanese Language and Culture – NPTEL (2025)
- Agile Scrum in Practice – Infosys (2024)
- Programming in Python – NMAMIT (2024)
- Time Management – Infosys (2024)

## WHY JAPAN ?

Japan's globally respected engineering culture, commitment to quality, and emphasis on precision strongly align with my career aspirations. I am particularly inspired by the Japanese principles of Monozukuri and Kaizen, which emphasize craftsmanship, continuous improvement, and attention to detail-values that I strive to apply in my technical work. Through my background in Artificial Intelligence, computer vision, and hands-on robotics and hardware projects, I have developed a strong interest in solving practical engineering problems and building technologies that can operate reliably in the real world. Japan's strength in robotics and advanced engineering makes it an environment where I believe I can learn from experienced engineers while contributing my background in AI and software. I am committed to learning Japanese, adapting to the professional culture, and continuously improving both my technical and communication skills. I see Japan not simply as a place to work, but as an environment where I can build a long-term career in robotics and AI while contributing to innovative, high-quality engineering systems.

## WHY TELEXISTENCE ?

Telexistence strongly aligns with my vision of building machines that expand what humanity is capable of. What particularly attracts me is the company's approach of combining robotics, AI, and teleoperation to create machines that can operate in the real world and extend human capabilities beyond physical limitations. My background in computer vision, deep learning, sensor fusion, and hands-on robotics has motivated me to pursue the development of intelligent machines that can perceive, learn, and act in physical environments. I see Telexistence as an ideal place to apply my existing skills, learn from multidisciplinary engineers, and contribute to the development of embodied AI and reliable robotic systems that can have a meaningful real-world impact.